



an notion of curvature.

displacement convex

generalizes convexity to curved spaces of measures

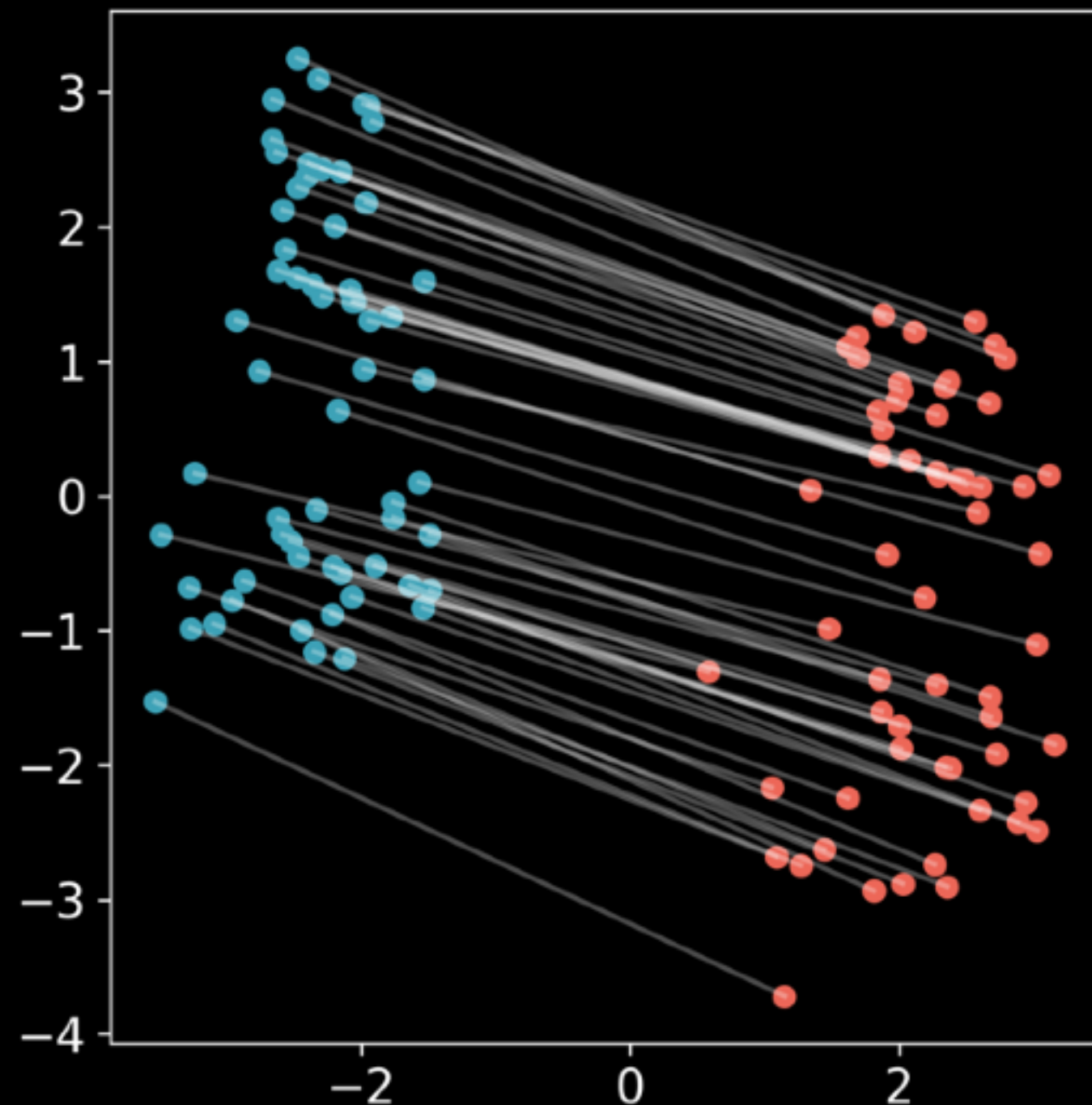
Geometric Properties

- Wasserstein space $\mathbb{W}_2$ is a Riemannian manifold (formal structure), *non-negative curvature*
- Convexity of functionals along Wasserstein geodesics provides **a notion of curvature**.
- **In Euclidean space:**
 - A functional $F : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex if:
$$F((1-t)x_0 + tx_1) \leq (1-t)F(x_0) + tF(x_1), \quad t \in [0,1].$$
 - Classical convexity along straight lines encodes flatness
- **In Wasserstein space:**
 - “Straight lines” are displacement interpolations between measures.
 - A functional $F : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$ is **displacement convex** if:
$$F(\rho_t) \leq (1-t)F(\rho_0) + tF(\rho_1), \quad t \in [0,1],$$
 whenever $(\rho_t)_{t \in [0,1]}$ is a Wasserstein geodesic.
 - This **generalizes convexity to curved spaces of measures**.
- Displacement convex functions are “nice” (more on that later). E.g., entropy, potential energy, etc

Intuition: Discrete/Particle View

Static OT

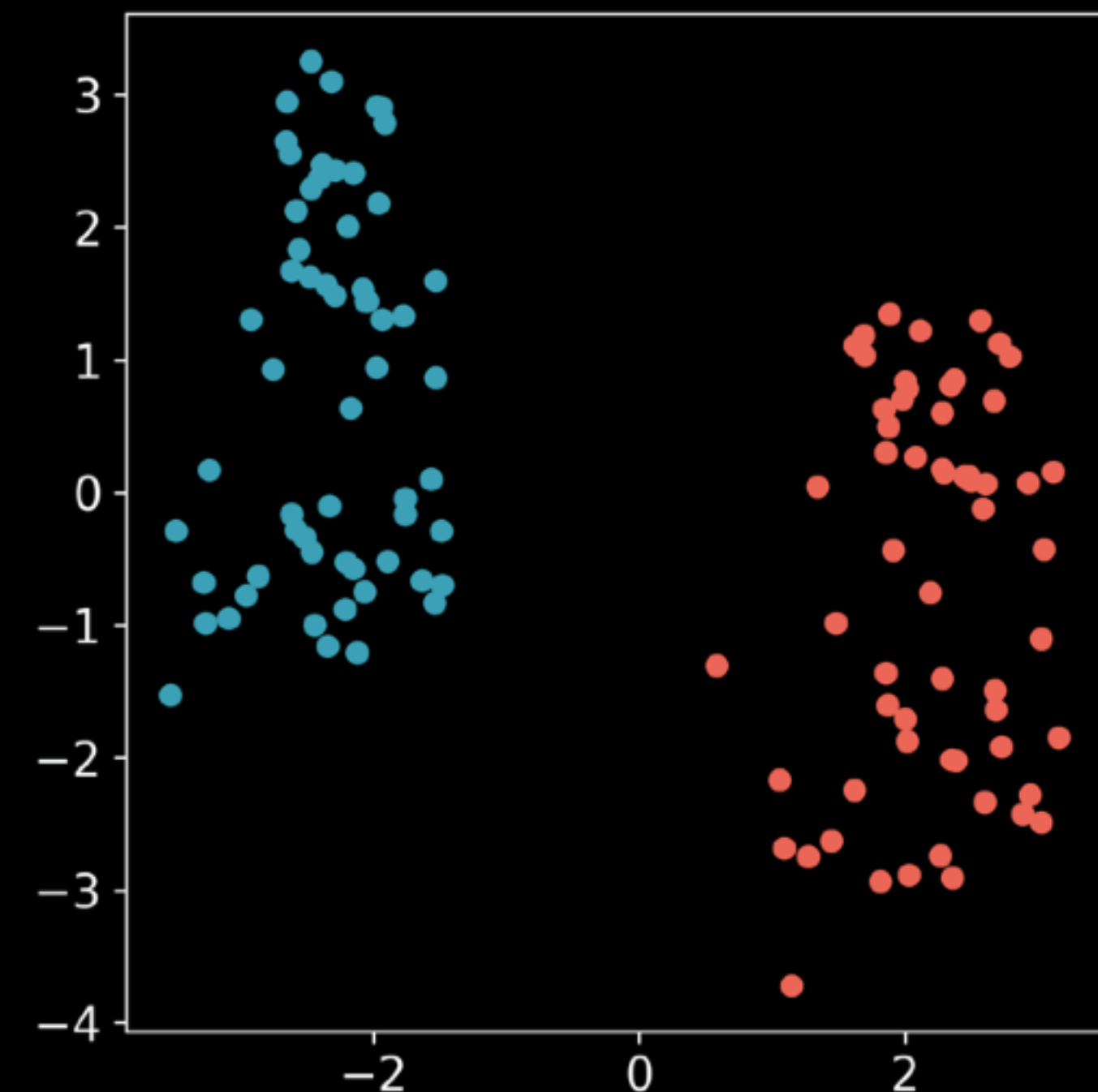
Assigns pairs (x, y)



Each particle matched with other particle(s);
ensemble of pairs defines joint distribution π

Dynamic OT

Describes trajectories $x \mapsto x_t$



Each particle follows straight line; ensemble
of particles evolves as ρ_t